

SIGMA: An Open-Source Interactive System for Mixed-Reality Task Assistance Research – Extended Abstract

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ABSTRACT

We introduce an open-source system called SIGMA (short for “Situ-ated Interactive Guidance, Monitoring, and Assistance”) as a plat-form for conducting research on task-assistive agents in mixed-reality scenarios. The system leverages the sensing and rendering affordances of a head-mounted mixed reality device in conjunction with large language and vision models to guide users step by step through procedural tasks. By open-sourcing the system, we aim to lower the barrier to entry, accelerate research in this space, and chart a path towards community-driven end-to-end evaluation of large language, vision, and multimodal models in the context of real-world interactive applications.

1 INTRODUCTION

Mixed-reality technologies promise to offer a powerful conduit for empowering individuals by providing just-in-time assistance in the physical world. Unsurprisingly, procedural task assistance has been one of the earliest envisioned applications for mixed reality [5] and has continued to attract significant attention [1, 3, 4, 6, 7].

Recent breakthroughs in generative AI and advances in large language, vision, and multimodal models have opened up new possibilities in this space. By providing a substrate of open-domain knowledge, inference, and generation capabilities, these models promise to help enable task assistance in open-ended (rather than predefined) scenarios. However, new challenges around how to use these models effectively also emerge; e.g., how to integrate them into real-time, streaming, egocentric settings; how to improve robustness and reduce hallucinations; how to mitigate safety risks, etc.

In this extended abstract we introduce **SIGMA**, shorthand for **S**ituated **I**nteractive **G**uidance, **M**onitoring and **A**ssistance, an open-source¹ empirical testbed system for research in mixed-reality task assistance. We provide an initial implementation that can leverage large language and large vision models to guide users through procedural tasks. The system relies on a client-server architecture where a HoloLens 2 device performs sensor capture and UI rendering, but perception and computation are offloaded live to a separate compute server, bypassing computational limits on current headset devices and fostering device-independent research and development. By open-sourcing the system, we aim to lower the barrier to entry and accelerate research at the intersection of mixed reality and AI.

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¹<https://aka.ms/psi-sigma>

2 SYSTEM OVERVIEW

We begin by introducing SIGMA’s core functionality via a sample interaction, shown in Figure 1. Interactions begin with a task identification sub-dialog where the system asks the user which task they need assistance with. The system can guide the user through known tasks with recipes that are manually defined in a task library. Alternatively, a second mode allows the system to automatically generate recipes for novel tasks via an LLM. In the latter mode, the system first uses an LLM prompt to generate a set of questions that elicit more information about the user’s context. After receiving the user’s responses to these questions, the system uses another LLM prompt to generate step-by-step instructions for the task at hand.

Once the task recipe is identified (or generated via the LLM), SIGMA begins to guide the user through the task. Throughout the interaction, the system displays the history of the dialog in a set of chat bubbles, and step-by-step instructions in a virtual task panel placed below the chat bubbles, as shown in Figure 1.B. The application currently supports linear tasks (no-branching) containing three types of steps: a *simple* step providing a single instruction to the user, a *complex* step that provides a high level instruction decomposing into a linear set of sub-steps (each with their own instructions), and a *gather* step that helps the user to find and collect a set of objects required for performing the task, such as tools, ingredients, etc.

For *simple* and *complex* steps, SIGMA provides the corresponding step instructions to the user, together with relevant holograms, and simply waits for the user to report completion of the step. Throughout *gather* steps, the system uses an open-world object detection model [8] to help the user find task-relevant objects. If one of the relevant objects is detected, SIGMA draws the user’s attention to it with a hologram animation and a virtual label with the object’s name above its location, as shown in Figure 1.F.

During the execution of *simple* and *complex* steps, SIGMA leverages the LLM to answer open-ended questions from the user. This feature is accomplished with another LLM prompt which has been designed to identify whether a user utterance is a question (as opposed to self-talk, a navigational command, etc.) and if so, generate a corresponding answer based on the context of the task and the current step. The resulting answer is displayed and spoken back to the user just like any other system utterance.

SIGMA is built using Platform for Situated Intelligence [2] (also known as \psi), an open-source framework that provides infrastructure, tools and components for developing applications that work with multimodal streaming data. The system is implemented using a client-server architecture. Specifically, a SIGMA client app runs on the mixed-reality headset and captures data from various sensors, including the RGB and depth camera; audio; and head, hand and gaze tracking information. The data is streamed live over a TCP/IP connection to a SIGMA server application that runs on a separate desktop computer. The server implements most of the compute functionality and streams back commands to the client app, which renders UI panels, holograms, and spatial audio accordingly.

The client-server architecture provides several important benefits for fast prototyping and research. First, the architecture enables

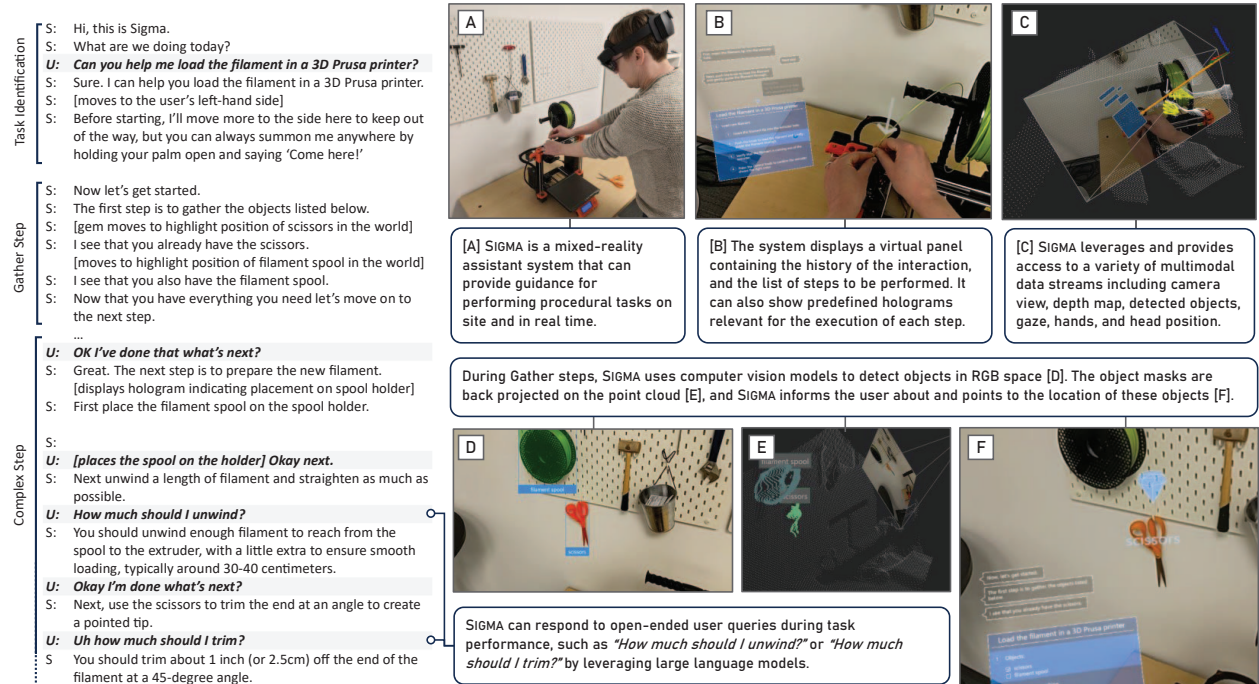


Figure 1: Trace from a sample interaction with SIGMA, with callouts illustrating various system features.

researchers to bypass existing computational limits on current head-set devices. It also enables faster debugging and development by leveraging the data replay capabilities of the underlying \psi framework. Finally, the architecture fosters device-independent research and development. While in the current implementation the client app targets the HoloLens 2 platform, new versions of the client app could be constructed and used to run on other devices (e.g., Android phones, Oculus headsets, etc.)

Researchers using SIGMA can leverage and benefit from the tools and affordances of the underlying \psi framework. \psi provides an efficient infrastructure for data logging. All the data streams captured or generated by the application can be automatically persisted to disk and the collected data can be easily inspected (offline or live) using Platform for Situated Intelligence Studio [2]. Additional information about the system, including architectural details, types of data collected, development tools, ethical and responsible AI development considerations are available at <https://aka.ms/psi-sigma>.

3 CONCLUSION

We have introduced SIGMA, an open-source end-to-end interactive system for mixed-reality procedural task assistance. The system harnesses the multimodal sensing and rendering capabilities of the HoloLens 2 device, together with large language and vision models to engage in mixed-initiative dialog and guide users step by step through procedural tasks. The system is built on a modern infrastructure with associated tools for working with temporally streaming data, and has a modular design that fosters research and exploration.

Overall, the set of features and the functionality in the current implementation provide a robust basis for future development and research. We hope to engage the research community and together extend SIGMA's capabilities and use it as a springboard for new research at the intersection of AI and mixed-reality task assistance.

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